LEDU_exercise2

Questions

Here is the code provided:

```
pacman::p_load(network, ergm, tnet, install = T)

consult <- Cross.Parker.Consulting.net.info
consult$w <- ifelse(consult$w < 4, 0, 1)
consult <- consult[consult$w == 1, ]
consult <- network(consult[, 1:2], directed = T)
consult %v% "gender" <- Cross.Parker.Consulting.node.gender
consult %v% "region" <- Cross.Parker.Consulting.node.region
consult %v% "orglevel" <- Cross.Parker.Consulting.node.orglevel
```

```
set.seed(510)
ergm.model <- ergm(consult ~ edges + mutual + gwesp(0.5, fixed = T) +
  nodematch("gender") + nodematch("region") +
  nodematch("orglevel") + nodeifactor("orglevel") +
  nodeofactor("orglevel") + absdiff("orglevel"), control = control.ergm(
    MCMC.samplesize = 5000,
    MCMC.burnin = 10000,
    MCMLE.maxit = 10
    ),
    verbose = T)
summary(ergm.model)
```

Here are the results provided for the questions:

```
Monte Carlo Maximum Likelihood Results:

                         Estimate Std. Error MCMC %    z value Pr(>|z|)
```

```
edges                   -4.064e+12  4.939e-01      0 -8.229e+12  < 1e-04 ***
mutual                   4.247e+00  5.186e-01      0  8.189e+00  < 1e-04 ***
gwesp.OTP.fixed.0.5      2.465e+12  2.995e-01      0  8.229e+12  < 1e-04 ***
nodematch.gender         1.404e-01  2.384e-01      1  5.890e-01 0.555892
nodematch.region        -9.278e-01  2.830e-01      1 -3.278e+00 0.001045 **
nodematch.orglevel       1.050e+00  4.258e-01      1  2.467e+00 0.013635 *
nodeifactor.orglevel.2   6.605e-01  8.150e-01      0  8.100e-01 0.417693
nodeifactor.orglevel.3   9.048e-01  8.049e-01      0  1.124e+00 0.260950
nodeifactor.orglevel.4   1.721e+00  7.317e-01      0  2.352e+00 0.018663 *
nodeifactor.orglevel.5   8.214e-01  1.077e+00      0  7.620e-01 0.445845
nodeofactor.orglevel.2   1.688e+00  8.093e-01      0  2.086e+00 0.037010 *
nodeofactor.orglevel.3   3.023e+00  8.392e-01      0  3.602e+00 0.000315 ***
nodeofactor.orglevel.4   2.141e+00  7.792e-01      0  2.748e+00 0.005992 **
nodeofactor.orglevel.5  -1.358e+00  1.139e+00      0 -1.193e+00 0.232937
absdiff.orglevel         5.002e-01  1.642e-01      1  3.045e+00 0.002323 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 2.274e+03  on 1640  degrees of freedom
 Residual Deviance: 4.548e+14  on 1625  degrees of freedom
```

Using the results provided, here are the answers to the questions:

```
•1: Based on visual inspection only, there seems to be a regional homophily between nodes, me
•2: The Monte Carlo Maximum Likelihood results are not significant for gender matching, meani
•3: The Monte Carlo Maximum Likelihood results are significant but negative for region matchi
•4: It is difficult to visually guess homophily or heterophily at an organizational level. Th
•5: Reading nodeifactor tests, vertices at the 4th organizational level seem to moderately at
•6: The nodeofactor() terms measures the effect of factor attributes on outward edges. In mor
•7: The absdiff() term measures the homophily based on continuous attributes instead of binar
```

On your own

Running the ERGM model on my own led to an error message (Error in ergm.MCMLE(init, s, s.obs, control = control, verbose = verbose, : Unconstrained MCMC sampling did not mix at all. Optimization cannot continue.). Writing the summary despite the message error shows unconvincing results:

```
# Monte Carlo Maximum Likelihood Results:

#   Estimate Std. Error MCMC %    z value Pr(>|z|)
```

```
#  edges                    5.651e+11  1.981e-02    100  2.852e+13   <1e-04 ***
#  mutual                  -2.241e+02  1.446e-02    100 -1.549e+04   <1e-04 ***
#  gwesp.OTP.fixed.0.5     -3.428e+11  1.202e-02    100 -2.852e+13   <1e-04 ***
#  nodematch.gender        -1.951e+01  1.351e-02    100 -1.444e+03   <1e-04 ***
#  nodematch.region        -2.320e+02  1.298e-02    100 -1.787e+04   <1e-04 ***
#  nodematch.orglevel      -2.632e+02  2.277e-02    100 -1.156e+04   <1e-04 ***
#  nodeifactor.orglevel.2   1.439e+02  2.534e-02    100  5.677e+03   <1e-04 ***
#  nodeifactor.orglevel.3   2.475e+01  2.542e-02    100  9.737e+02   <1e-04 ***
#  nodeifactor.orglevel.4  -4.949e+01  2.373e-02    100 -2.085e+03   <1e-04 ***
#  nodeifactor.orglevel.5   1.335e+02  2.980e-02    100  4.482e+03   <1e-04 ***
#  nodeofactor.orglevel.2  -1.129e+02  2.456e-02    100 -4.595e+03   <1e-04 ***
#  nodeofactor.orglevel.3  -9.666e+01  2.417e-02    100 -4.000e+03   <1e-04 ***
#  nodeofactor.orglevel.4  -2.810e+02  2.222e-02    100 -1.265e+04   <1e-04 ***
#  nodeofactor.orglevel.5   5.940e+02  2.745e-02    100  2.164e+04   <1e-04 ***
#  absdiff.orglevel        -6.109e+01  1.038e-02    100 -5.886e+03   <1e-04 ***
#  ---
#  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Null Deviance: 2.274e+03  on 1640  degrees of freedom
# Residual Deviance: 4.479e+14  on 1625  degrees of freedom
#
# AIC: 4.479e+14  BIC: 4.479e+14  (Smaller is better. MC Std. Err. = 5.98e+11)
# Warnings:
#
#  * The following linear dependence has been detected among the model statistics and/or esti
#
#  16568/10049 * edges = gwesp.OTP.fixed.0.5
```

Using a simpler model, removing gwesp specification showed new & different results:

```
set.seed(510)
ergm.model <- ergm(consult ~ edges + mutual +
                     nodematch("gender") + nodematch("region") +
                     nodematch("orglevel") + nodeifactor("orglevel") +
                     nodeofactor("orglevel") + absdiff("orglevel"), control = control.ergm(
                       MCMC.samplesize = 5000,
                       MCMC.burnin = 10000,
                       MCMLE.maxit = 10
                     ),
                   verbose = T)
summary(ergm.model)
```

```
Monte Carlo Maximum Likelihood Results:

                      Estimate Std. Error MCMC % z value Pr(>|z|)
edges                 -3.27777    0.47170      0  -6.949   <1e-04 ***
mutual                 3.94090    0.31673      0  12.442   <1e-04 ***
nodematch.gender      -0.05306    0.13973      0  -0.380   0.7042
nodematch.region       0.13823    0.12806      0   1.079   0.2804
nodematch.orglevel    -0.28713    0.24681      0  -1.163   0.2447
nodeifactor.orglevel.2  0.87747    0.42041      0   2.087   0.0369 *
nodeifactor.orglevel.3  0.60088    0.40895      0   1.469   0.1417
nodeifactor.orglevel.4  0.60948    0.37956      0   1.606   0.1083
nodeifactor.orglevel.5  0.55657    0.51831      0   1.074   0.2829
nodeofactor.orglevel.2 -0.94591    0.38738      0  -2.442   0.0146 *
nodeofactor.orglevel.3 -0.47655    0.37009      0  -1.288   0.1979
nodeofactor.orglevel.4 -0.27701    0.34061      0  -0.813   0.4161
nodeofactor.orglevel.5 -0.44775    0.47656      0  -0.940   0.3474
absdiff.orglevel      -0.07782    0.11385      0  -0.684   0.4943
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 2273.5  on 1640  degrees of freedom
 Residual Deviance:  880.1  on 1626  degrees of freedom

AIC: 908.1  BIC: 983.8  (Smaller is better. MC Std. Err. = 0.6431)
```

```
These new results contredict previous analysis: only one organizational level significantly a
```